

TG ECET (WP)-2026**SYLLABUS FOR ELECTRICAL AND ELECTRONICS ENGINEERING****MATHEMATICS (10 Marks)**

S.No	Syllabus
1	UNIT I: MATRICES AND DETERMINANTS Definition of Matrix-order of a matrix-Type of matrices - Algebra of matrices- sum, difference, scalar multiplication and product of 2x2 Matrix - Equality of two matrices- Transpose of a Matrix – Determinant of a square Matrix of order 2×2 .
2	UNIT II: CO-ORDINATE GEOMETRY Straight lines: Slope of straight line - Various forms of straight line: Point-Slope form, Two- Point form, Intercept form, and General equation of a Straight-line Conditions for Parallelism and Perpendicularity of the Straight lines – Equation of a straight line parallel and perpendicular to the given straight line -Distance between two Parallel lines.
3	UNIT III: TRIGONOMETRY Trigonometric Ratios of Compound angles (without proof) - Trigonometric Ratios of Multiple and Sub- multiple angles ($2A$, $3A$, $A/2$).

ELECTRICAL AND ELECTRONICS ENGINEERING (90 Marks)

S.No	Syllabus
1	UNIT I: BASIC ELECTRICAL ENGINEERING Ohms Law , Kirchoff's Laws, Series and Parallel resistance circuits, star/delta transformation, Network theorems, Work, Power and Energy. Heating effects of Electric current, Magnetic effects of Electric current, Electromagnetic Induction, Types of Electrical Engineering materials-Conducting, Semi conducting, Magnetic, Insulating, Dielectric and Special purpose materials- Properties and uses. Batteries-Types, Properties, applications, charging methods and efficiency calculation.
2	UNIT II: ELECTRICAL & ELECTRONIC MEASURING INSTRUMENTS Measuring Instruments: Classification, Deflection, Controlling and damping torques - Construction, Working Principle of Operation of moving Coil, Moving Iron, Dynamometer type, Induction type meters, Induction type Energy meter, Errors. Instrument Transformers, Shunts and Multipliers. Measurement of Resistance, construction and working of Megger, Potentiometer, Transducers and Sensors – Types, Thermistors, Thermocouple, LVDT and Strain gauges. Electronics and digital instruments. Rectifier type, Digital Multimeter, Digital Energy meter, Digital Frequency meter and Clamp meter.

3	UNIT III: D.C. MACHINES D.C. Generators: Construction, Operation, Classification of DC Generators- EMF Equation, Types of Armature Windings, Armature reaction, Characteristics, Calculation of Efficiency, Parallel operation and Applications. DC Motors: Principle of operation, Back EMF, Torque Equation, Types, Characteristics, Starters, Speed Control methods, Testing, Losses, Efficiency and Applications. Types of Electrical braking systems.
4	UNIT IV: A.C. CIRCUITS A.C. Circuits: Fundamentals, Average value, RMS value, Form factor and Peak factor of periodic wave forms. Single phase AC Series and parallel Circuits, R-L-C Resonant circuits. Polyphase Circuits, Measurement of 3-phase power by 1, 2, 3 Wattmeter's method for balanced loads.
5	UNIT V: TRANSFORMERS Transformers: Single-phase Transformers - Construction, Operation, EMF Equation, Testing, Equivalent circuit, Regulation, All day efficiency and Applications. Auto-transformers, Three-phase transformers, types of connections and their applications. Differences between Distribution and Power Transformers.
6	UNIT VI: A.C. MACHINES Three-Phase Induction Motors: Construction, Principle of Operation, Torque Equation, torque-slip characteristics, losses, efficiency, testing, circle diagram, speed control methods, starters and applications, performance of double-cage Induction motors. Single-phase Induction Motors and Special purpose Motors: Types, Principle of operation, applications. Single Phase induction motors. Principle of operation and applications of AC series motors, Universal Motor, Stepper Motor, Servomotor and BLDC Motors. Alternators: Construction, Operation, EMF equation, regulation, testing and parallel Operation. Synchronous Motors: Construction, Operation and performance, effect of field excitation, 'V'-Curve and inverted 'V'- Curve, methods of Starting and applications of Synchronous motors. Hunting and its effects.
7	UNIT VII:-ELECTRICAL POWER SYSTEMS Generating Stations: Conventional and Non-conventional sources of energy. Working principle and classification of Thermal, Hydel, Nuclear and Gas Power stations. Components and auxiliaries of power stations. Renewable energy sources; Solar, Photovoltaic and windmill. Transmission and Distribution: Types of supply systems, Transmission line parameters, inductance and capacitance, performance of short and medium lines, regulation, Ferranti effect, Corona, Skin effect. Components of Transmission lines, supports, conductor

	spacing, ground clearance and sag. Line Insulators, voltage distribution across the string, string efficiency, methods of improving string efficiency. Types of Substations and Substation equipment. Merits and demerits of Gas Insulated Substation. HVDC Transmission: Basic concepts, types, Advantages and Disadvantages. Distribution: Radial and ring distributors, variation of load voltage, micro and smart grids.
8	UNIT VIII: - PROTECTION OF POWER SYSTEMS Switch gear – Types of faults in power system. Principle of operation, Types and Uses of Switches, Fuses, Isolators, Circuit Breakers and Current Limiting reactors. Relays: Basic requirements, Classification, Principle of Operation of Electromagnetic type and Induction type over current relay, Directional over current relays, PSM and TSM settings. Principle of impedance distance relays. Protection of alternators, Transformers, Bus-bars, Transmission lines and feeders. Types of voltage surges, Lightning arrestors, neutral grounding. Basic concepts of Digital Protection.
9	UNIT IX: - ELECTRICAL ESTIMATION AND UTILISATION Electric Wiring: Tools, Wires, Types of wiring systems, Accessories, wiring systems Lamp Circuits. Estimating and costing of domestic, industrial, power, irrigation pump sets, overhead lines and 11 kV Substations. Rural electrification, departmental tests, Earthing, maintenance of electrical machines. Electric Lighting: Requirements of good lighting, different types of lamp fittings, laws of illumination. Electric heating: Requirements of good heating element, Types of resistance heating, Electric arc furnaces, Types of induction heating, Dielectric heating-Principle of operation and Applications. Electric welding: Types, working of welding, welding generator and welding transformer, Applications. Electric Traction: Systems of Train Electrification, Speed-time Curves (Trapezoidal speed time curve only) for different services, Schedule speed, Tractive Effort, coefficient of adhesion, Traction system auxiliaries, Traction motors, Supply systems.
10	UNIT X: - BASIC ELECTRONICS Classification of resistors, inductors, capacitors and applications. Semi - conductor Devices: Working and Characteristics of PN junction diode and Zener diode. Transistors- PNP and NPN Transistors, characteristics, configurations and biasing methods. Half and Full wave rectifiers, Bridge rectifiers, Filters, Zener diode regulation. Special devices: UJT, FET, JFET, MOSFET. Amplifiers- Types, Principles of operation, Characteristics, Multistage, Coupled amplifiers, Power amplifiers, and Feedback amplifiers. Oscillators- Types, working, principle of operation and applications. Non linear wave shaping circuits, operational amplifiers, differential amplifiers, Tuned amplifiers characteristics, block diagram, types. Pin configuration of 741 IC.

11	UNIT XI: - DIGITAL ELECTRONICS Digital Electronics- Various number systems, inter conversions, Boolean Algebra, Logic families, performance of AND, OR, NOT, NOR, NAND, EX-OR gates, combinational Logic Circuits, sequential logic circuits, Registers, A/D and D/A converters, counters, timers and flip-flops, memories.
12	UNIT XII: - POWER ELECTRONICS Power Electronic Devices: Construction and working of SCR, DIAC, TRIAC, LASCR, IGBT. Volt- ampere characteristics, Triggering of SCR using UJT and Protection of SCR. Converters- Rectifiers, Choppers, Inverters, AC regulators and Cyclo converters. Classification, Block Diagram and applications of UPS, SMPS. Applications of Power Electronics circuits.
13	UNIT XIII. PLC AND C LANGUAGE PLC- Components and concepts of Ladder diagrams, applications. C Language-key words & Identifiers, constants & variables, Operators & Expressions- basic programming.
14	UNIT XIV. ELECTRICAL VEHICLES Basic concept of Electric Vehicle (EV) – Hybrid Electric Vehicles (HEV) – Plug-in Hybrid Electric Vehicles – types of EV based on Drive Train configuration – various components in EV.